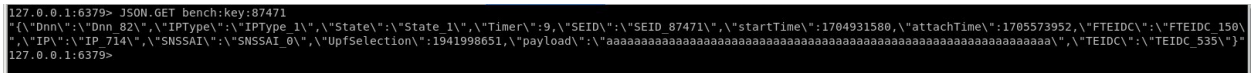
Báo cáo kết quả benchmark của redis-cluster và aerospike

Báo cáo kết quả benchmark của redis-cluster và aerospike 1
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1, Dataset

a, Redis


127.0.0.1:6379> JSON.GET bench:key:87471
"{\"Dnn\":\"Dnn_82\",\"IPType\":\"IPType_1\",\"State\":\"State_1\",\"Timer\":9,\"SEID\":\"SEID_87471\",\"startTime\":1704931580,\"attachTime\":1705573952,\"FTEIDC\":\"FTEIDC_150\
",\"IP\":\"IP_714\",\"SNSSAI\":\"SNSSAI_0\",\"UpfSelection\":1941998651,\"payload\":\"aaaa

Command:

- **Redis:**

FT.CREATE idx ON JSON PREFIX 1 bench:key: SCHEMA $.Dnn AS Dnn TAG $.IPType AS IPType TAG $.State AS State TAG $.Timer AS Timer NUMERIC $.SEID AS SEID TAG $.startTime AS startTime NUMERIC $.attachTime AS attachTime NUMERIC $.FTEIDC AS FTEIDC TAG $.IP AS IP TAG $.SNSSAI AS SNSSAI TAG $.UpfSelection AS UpfSelection NUMERIC $.TEIDC AS TEIDC TAG

```
err := client.Do(ctx, "FT.CREATE", "idx", "ON", "JSON", "PREFIX", "1", "bench:key:", "SCHEMA",
  "$.Dnn", "AS", "Dnn", "TAG",
  "$.IPType", "AS", "IPType", "TAG",
  "$.State", "AS", "State", "TAG",
  "$.Timer", "AS", "Timer", "NUMERIC",
  "$.SEID", "AS", "SEID", "TAG",
  "$.startTime", "AS", "startTime", "NUMERIC",
  "$.attachTime", "AS", "attachTime", "NUMERIC",
  "$.FTEIDC", "AS", "FTEIDC", "TAG",
  "$.IP", "AS", "IP", "TAG",
  "$.SNSSAI", "AS", "SNSSAI", "TAG",
  "$.UpfSelection", "AS", "UpfSelection", "NUMERIC",
  "$.TEIDC", "AS", "TEIDC", "TAG").Err()
```

- **Aerospike:**

CREATE STRING INDEX idx_Dnn ON smf.benchmark_session (Dnn);

CREATE STRING INDEX idx_IPType ON smf.benchmark_session (IPType);

CREATE STRING INDEX idx_State ON smf.benchmark_session (State);

CREATE NUMERIC INDEX idx_Timer ON smf.benchmark_session (Timer);

CREATE STRING INDEX idx_SEID ON smf.benchmark_session (SEID);

CREATE NUMERIC INDEX idx_startTime ON smf.benchmark_session (startTime);

CREATE NUMERIC INDEX idx_attachTime ON smf.benchmark_session (attachTime);

CREATE STRING INDEX idx_FTEIDC ON smf.benchmark_session (FTEIDC);

CREATE STRING INDEX idx_IP ON smf.benchmark_session (IP);

CREATE STRING INDEX idx_SNSSAI ON smf.benchmark_session (SNSSAI);

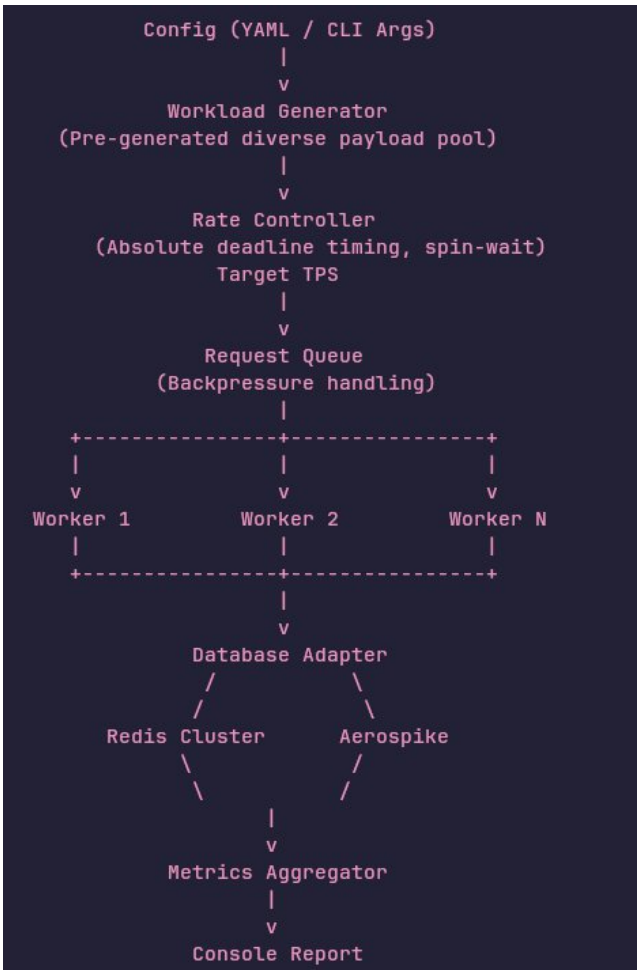
CREATE NUMERIC INDEX idx_UpfSelection ON smf.benchmark_session (UpfSelection);

CREATE STRING INDEX idx_TEIDC ON smf.benchmark_session (TEIDC);

```
indexes := map[string]as.IndexType{
  "Dnn":          as.STRING,
  "IPType":       as.STRING,
  "State":        as.STRING,
  "Timer":        as.NUMERIC,
  "SEID":         as.STRING,
  "startTime":    as.NUMERIC,
  "attachTime":   as.NUMERIC,
  "FTEIDC":       as.STRING,
  "IP":           as.STRING,
  "SNSSAI":       as.STRING,
  "UpfSelection": as.NUMERIC,
  "TEIDC":        as.STRING,
}

for binName, idxType := range indexes {
  idxName := "idx_" + binName
  task, err := a.client.CreateIndex(nil, a.namespace, a.setName, idxName, binName, idxType)
  if err != nil {
    if strings.Contains(err.Error(), "Index already exists") || strings.Contains(err.Error(), "already exists") {
      continue
    }
    return err
  }
  if task != nil {
    <-task.OnComplete()
  }
}
```

2, Kiến trúc hệ thống benchmark



==========

Benchmark Results

==========

Database: redis

Measurement: 10.003s

Total Duration: 10.045s

Target TPS: 10000

Issued TPS: 9998.50 <-- Tốc độ thực tế phát request

Completed TPS: 9853.30

Success TPS: 9853.30

Error TPS: 0.00

Total Issued: 99985

Total Completed: 98533

Success: 98533

Errors: 0

Dropped (queue): 1400   <-- Số request bị vứt đi do hàng đợi đầy

Abandoned: 52   <-- Request còn kẹt chưa xử lý xong sau Grace Period

Latency:

  Avg: 1.23ms

  P50:   812µs

  P95:   3.91ms

  P99:   9.27ms

  Max: 45.52ms

```

- **Target TPS vs Issued TPS**: Nếu máy benchmark theo kịp, 2 con số này phải gần bằng nhau.

- **Latency (P50, P95, P99)**: Phản ánh thời gian thực tế để gọi DB (không tính thời gian nằm trong queue của công cụ test). P99 = 9.27ms nghĩa là 99% request có thời gian xử lý dưới mức này.

- **Dropped**: Lượng request chủ động bị drop vì hàng đợi đã đầy. Đây là dấu hiệu cơ sở dữ liệu đã bị quá tải (saturated).
```

3, Kết quả benchmark

a, Triển khai 1:

- **2 pod Aerospike: (tổng 8 vcpu, 36 Gi)**

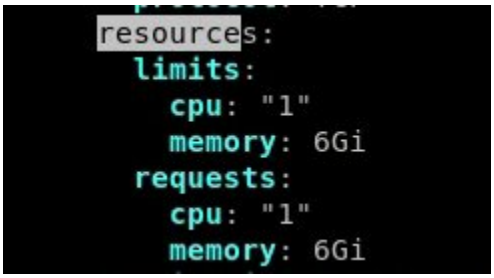


- **Cụm redis-cluster gồm 3 master 3 slave: (tổng 9 vcpu, 36Gi)**

MASTER pod: (tổng 6 vcpu, 18Gi)



SLAVE pod: (tổng 3 vcpu, 18Gi)



=> Kết quả theo từng loại truy vấn

1, SET

```
                                                                Autoscroll:On      FullScreen:
==========================================
          Benchmark Results
==========================================

Database:          aerospike
Measurement:       10s
Total Duration:    10s

Target TPS:        20000
Issued TPS:        19999.78
Completed TPS:     19999.21
Success TPS:       19999.21
Error TPS:         0.00

Total Issued:      199999
Total Completed:   199999
Success:           199999
Errors:            0
Error Rate:        0.00%

Overall Latency:
  Avg:             367.065µs
  P50:             345.856µs
  P95:             441.489µs
  P99:             586.243µs
  Max:             12.059615ms

Per-Operation Latency:
  SET (199999 requests):
    Avg: 367.065µs, P50: 345.856µs, P95: 441.489µs, P99: 586.243µs, Max: 12.059615ms

==========================================
```

SET AEROSPIKE

```
==========================================
          Benchmark Results
==========================================

Database:          redis
Measurement:       10s
Total Duration:    10.049s

Target TPS:        20000
Issued TPS:        19999.68
Completed TPS:     19902.80
Success TPS:       19902.80
Error TPS:         0.00

Total Issued:      199998
Total Completed:   199998
Success:           199998
Errors:            0
Error Rate:        0.00%

Overall Latency:
  Avg:             212.172µs
  P50:             154.558µs
  P95:             404.518µs
  P99:             1.029941ms
  Max:             21.851417ms

Per-Operation Latency:
  SET (199998 requests):
    Avg: 212.172µs, P50: 154.558µs, P95: 404.518µs, P99: 1.029941ms, Max: 21.851417ms

==========================================
```

SET REDIS

2, GET

```
==========================================
          Benchmark Results
==========================================

Database:          aerospike
Measurement:       10s
Total Duration:    10s

Target TPS:        100000
Issued TPS:        99998.98
Completed TPS:     99998.23
Success TPS:       99998.23
Error TPS:         0.00

Total Issued:      999992
Total Completed:   999992
Success:           999992
Errors:            0
Error Rate:        0.00%

Overall Latency:
  Avg:             87.153µs
  P50:             80.978µs
  P95:             120.669µs
  P99:             177.1µs
  Max:             10.9549ms

Per-Operation Latency:
  GET (999992 requests):
    Avg: 87.153µs, P50: 80.978µs, P95: 120.669µs, P99: 177.1µs, Max: 10.9549ms
```

GET AEROSPIKE

```
==========================================
          Benchmark Results
==========================================

Database:          redis
Measurement:       10s
Total Duration:    11.284s

Target TPS:        90000
Issued TPS:        89999.74
Completed TPS:     79756.94
Success TPS:       79756.94
Error TPS:         0.00

Total Issued:      900001
Total Completed:   900001
Success:           900001
Errors:            0
Error Rate:        0.00%

Overall Latency:
  Avg:             124.868µs
  P50:             116.698µs
  P95:             174.555µs
  P99:             240.282µs
  Max:             9.594232ms

Per-Operation Latency:
  GET (900001 requests):
    Avg: 124.868µs, P50: 116.698µs, P95: 174.555µs, P99: 240.282µs, Max: 9.594232ms

==========================================
```

GET REDIS

3, DELETE

```
==========================================
          Benchmark Results
==========================================

Database:          aerospike
Measurement:       10s
Total Duration:    10s

Target TPS:        100000
Issued TPS:        99999.28
Completed TPS:     99997.33
Success TPS:       99997.33
Error TPS:         0.00

Total Issued:      999998
Total Completed:   999998
Success:           999998
Errors:            0
Error Rate:        0.00%

Overall Latency:
  Avg:             77.535µs
  P50:             71.012µs
  P95:             115.208µs
  P99:             178.313µs
  Max:             9.510366ms

Per-Operation Latency:
  DELETE (999998 requests):
    Avg: 77.535µs, P50: 71.012µs, P95: 115.208µs, P99: 178.313µs, Max: 9.510366ms

==========================================
```

DELETE AEROSPIKE

```
==========================================
          Benchmark Results
==========================================

Database:          redis
Measurement:       10s
Total Duration:    10.66s

Target TPS:        100000
Issued TPS:        99999.34
Completed TPS:     93805.02
Success TPS:       93805.02
Error TPS:         0.00

Total Issued:      999996
Total Completed:   999996
Success:           999996
Errors:            0
Error Rate:        0.00%

Overall Latency:
  Avg:             106.123µs
  P50:             99.824µs
  P95:             141.626µs
  P99:             192.478µs
  Max:             6.258571ms

Per-Operation Latency:
  DELETE (999996 requests):
    Avg: 106.123µs, P50: 99.824µs, P95: 141.626µs, P99: 192.478µs, Max: 6.258571ms

==========================================
```

DELETE REDIS

4, EQUALITY QUERY

```
==========================================
          Benchmark Results
==========================================

Database:         aerospike
Measurement:      10s
Total Duration:   13.698s

Target TPS:       3000
Issued TPS:       2999.89
Completed TPS:    2190.01
Success TPS:      2190.01
Error TPS:        0.00

Total Issued:     29999
Total Completed:  29999
Success:          29999
Errors:           0
Error Rate:       0.00%

Overall Latency:
  Avg:            4.562291ms
  P50:            4.545167ms
  P95:            6.994188ms
  P99:            8.264681ms
  Max:            16.309579ms

Per-Operation Latency:
  QUERY_TAG (29999 requests):
    Avg: 4.562291ms, P50: 4.545167ms, P95: 6.994188ms, P99: 8.264681ms, Max: 16.309579ms

==========================================
```

TAG AEROSPIKE

```
==========================================
          Benchmark Results
==========================================

Database:         redis
Measurement:      10s
Total Duration:   13.576s

Target TPS:       3000
Issued TPS:       2999.61
Completed TPS:    2209.56
Success TPS:      2209.56
Error TPS:        0.00

Total Issued:     29997
Total Completed:  29997
Success:          29997
Errors:           0
Error Rate:       0.00%

Overall Latency:
  Avg:            4.523488ms
  P50:            3.323286ms
  P95:            8.047009ms
  P99:            13.600295ms
  Max:            287.977265ms

Per-Operation Latency:
  QUERY_TAG (29997 requests):
    Avg: 4.523488ms, P50: 3.323286ms, P95: 8.047009ms, P99: 13.600295ms, Max: 287.977265ms

==========================================
```

TAG REDIS

5, RANGE QUERY

```
==========================================
          Benchmark Results
==========================================

Database:          aerospike
Measurement:       10.001s
Total Duration:    10.569s

Target TPS:        600
Issued TPS:        599.86
Completed TPS:     567.62
Success TPS:       567.62
Error TPS:         0.00

Total Issued:      5999
Total Completed:   5999
Success:           5999
Errors:            0
Error Rate:        0.00%

Overall Latency:
  Avg:             17.172529ms
  P50:             2.162481ms
  P95:             59.50676ms
  P99:             72.362821ms
  Max:             86.559685ms

Per-Operation Latency:
  QUERY_NUMERIC (5999 requests):
    Avg: 17.172529ms, P50: 2.162481ms, P95: 59.50676ms, P99: 72.362821ms, Max: 86.559685ms

==========================================
```

NUMERIC AEROSPIKE

```
==========================================
          Benchmark Results
==========================================

Database:          redis
Measurement:       10s
Total Duration:    10.003s

Target TPS:        700
Issued TPS:        699.98
Completed TPS:     699.82
Success TPS:       699.82
Error TPS:         0.00

Total Issued:      7000
Total Completed:   7000
Success:           7000
Errors:            0
Error Rate:        0.00%

Overall Latency:
  Avg:             12.631187ms
  P50:             9.749233ms
  P95:             19.565371ms
  P99:             137.021703ms
  Max:             344.731491ms

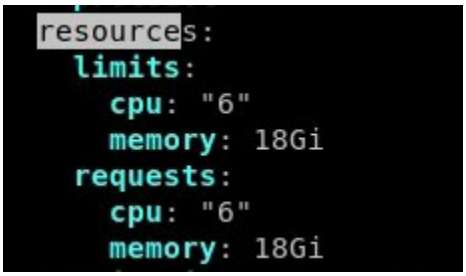
Per-Operation Latency:
  QUERY_NUMERIC (7000 requests):
    Avg: 12.631187ms, P50: 9.749233ms, P95: 19.565371ms, P99: 137.021703ms, Max: 344.731491ms

==========================================
stream closed EOF for hlcpc05/bmredis-runner-spw2h (benchmark)
```

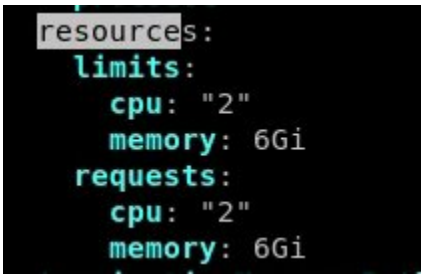
NUMERIC REDIS

b, Triển khai 2:

- 2 pods Aerospike (tổng 12 vcpu, 36Gi)



- 6 pods Redis, 3 pods Master, 3 pods Slave (tổng 12 vcpu, 36Gi)



=> Sau khi nâng cấp tài nguyên thì redis nhanh hơn đáng kể, aerospike gần như không thay đổi gì

```
Database:         aerospike
Measurement:      10s
Total Duration:   10s

Target TPS:       20000
Issued TPS:       19999.71
Completed TPS:    19999.17
Success TPS:      19999.17
Error TPS:        0.00

Total Issued:     199998
Total Completed:  199998
Success:          199998
Errors:           0
Error Rate:       0.00%

Overall Latency:
  Avg:            349.222µs
  P50:            328.77µs
  P95:            425.244µs
  P99:            595.476µs
  Max:            8.945476ms

Per-Operation Latency:
  SET (199998 requests):
    Avg: 349.222µs, P50: 328.77µs, P95: 425.244µs, P99: 595.476µs, Max: 8.945476ms
```

SET AEROSPIKE

```
Database:         redis
Measurement:      10s
Total Duration:   10s

Target TPS:       20000
Issued TPS:       19999.75
Completed TPS:    19999.55
Success TPS:      19999.55
Error TPS:        0.00

Total Issued:     199998
Total Completed:  199998
Success:          199998
Errors:           0
Error Rate:       0.00%

Overall Latency:
  Avg:            168.846µs
  P50:            153.59µs
  P95:            215.905µs
  P99:            366.032µs
  Max:            11.029245ms

Per-Operation Latency:
  SET (199998 requests):
    Avg: 168.846µs, P50: 153.59µs, P95: 215.905µs, P99: 366.032µs, Max: 11.029245ms
```

SET REDIS

```
Database:         aerospike
Measurement:      10s
Total Duration:   10s

Target TPS:       10000
Issued TPS:       9999.82
Completed TPS:    9999.81
Success TPS:      9999.81
Error TPS:        0.00

Total Issued:     99999
Total Completed:  99999
Success:          99999
Errors:           0
Error Rate:       0.00%

Overall Latency:
  Avg:            92.08µs
  P50:            85.635µs
  P95:            116.728µs
  P99:            173.015µs
  Max:            10.322195ms

Per-Operation Latency:
  GET (99999 requests):
    Avg: 92.08µs, P50: 85.635µs, P95: 116.728µs, P99: 173.015µs, Max: 10.322195ms
```

GET AEROSPIKE

```
Database:         redis
Measurement:      10s
Total Duration:   10s

Target TPS:       10000
Issued TPS:       9999.84
Completed TPS:    9999.82
Success TPS:      9999.82
Error TPS:        0.00

Total Issued:     99999
Total Completed:  99999
Success:          99999
Errors:           0
Error Rate:       0.00%

Overall Latency:
  Avg:            99.971µs
  P50:            94.901µs
  P95:            118.507µs
  P99:            152.026µs
  Max:            4.180631ms

Per-Operation Latency:
  GET (99999 requests):
    Avg: 99.971µs, P50: 94.901µs, P95: 118.507µs, P99: 152.026µs, Max: 4.180631ms
```

GET REDIS

```
Database:         aerospike
Measurement:      10s
Total Duration:   10s

Target TPS:       100000
Issued TPS:       99998.88
Completed TPS:    99998.28
Success TPS:      99998.28
Error TPS:        0.00

Total Issued:     999990
Total Completed:  999990
Success:          999990
Errors:           0
Error Rate:       0.00%

Overall Latency:
  Avg:            73.864µs
  P50:            68.663µs
  P95:            107.399µs
  P99:            157.978µs
  Max:            10.222251ms

Per-Operation Latency:
  DELETE (999990 requests):
    Avg: 73.864µs, P50: 68.663µs, P95: 107.399µs, P99: 157.978µs, Max: 10.222251ms
```

DELETE AEROSPIKE

```
Database:         redis
Measurement:      10s
Total Duration:   10.615s

Target TPS:       100000
Issued TPS:       99999.05
Completed TPS:    94206.76
Success TPS:      94206.76
Error TPS:        0.00

Total Issued:     999992
Total Completed:  999992
Success:          999992
Errors:           0
Error Rate:       0.00%

Overall Latency:
  Avg:            105.542µs
  P50:            99.227µs
  P95:            140.205µs
  P99:            184.22µs
  Max:            9.249804ms

Per-Operation Latency:
  DELETE (999992 requests):
    Avg: 105.542µs, P50: 99.227µs, P95: 140.205µs, P99: 184.22µs, Max: 9.249804ms
```

DELETE REDIS

```
Database:         aerospike
Measurement:      10s
Total Duration:   15.008s

Target TPS:       3300
Issued TPS:       3299.89
Completed TPS:    2127.05
Success TPS:      2127.05
Error TPS:        0.00

Total Issued:     32999
Total Completed:  31923
Success:          31923
Errors:           0
Abandoned:        1076
Error Rate:       0.00%

Overall Latency:
  Avg:            4.698552ms
  P50:            4.647389ms
  P95:            7.841622ms
  P99:            9.560527ms
  Max:            16.178947ms

Per-Operation Latency:
  QUERY_TAG (31923 requests):
    Avg: 4.698552ms, P50: 4.647389ms, P95: 7.841622ms, P99: 9.560527ms, Max: 16.178947ms
```

TAG AEROSPIKE

```
Database:         redis
Measurement:      10s
Total Duration:   15.081s

Target TPS:       3700
Issued TPS:       3699.94
Completed TPS:    2394.73
Success TPS:      2394.73
Error TPS:        0.00

Total Issued:     37001
Total Completed:  36114
Success:          36114
Errors:           0
Abandoned:        887
Error Rate:       0.00%

Overall Latency:
  Avg:            4.172986ms
  P50:            3.209931ms
  P95:            7.462504ms
  P99:            12.186772ms
  Max:            362.491703ms

Per-Operation Latency:
  QUERY_TAG (36114 requests):
    Avg: 4.172986ms, P50: 3.209931ms, P95: 7.462504ms, P99: 12.186772ms, Max: 362.491703ms
```

TAG REDIS

```
Database:         aerospike
Measurement:      10s
Total Duration:   15.092s

Target TPS:       700
Issued TPS:       699.90
Completed TPS:    303.07
Success TPS:      303.07
Error TPS:        0.00

Total Issued:     6999
Total Completed:  4574
Success:          4574
Errors:           0
Abandoned:        2425
Error Rate:       0.00%

Overall Latency:
  Avg:            32.854608ms
  P50:            2.218589ms
  P95:            102.348781ms
  P99:            109.94265ms
  Max:            130.407976ms

Per-Operation Latency:
  QUERY_NUMERIC (4574 requests):
    Avg: 32.854608ms, P50: 2.218589ms, P95: 102.348781ms, P99: 109.94265ms, Max: 130.407976ms
```

NUMERIC AEROSPIKE

```
Database:         redis
Measurement:      10.001s
Total Duration:   11.76s

Target TPS:       3000
Issued TPS:       2999.58
Completed TPS:    2550.86
Success TPS:      2550.86
Error TPS:        0.00

Total Issued:     29998
Total Completed:  29998
Success:          29998
Errors:           0
Error Rate:       0.00%

Overall Latency:
  Avg:            3.819567ms
  P50:            2.840784ms
  P95:            7.126519ms
  P99:            11.679857ms
  Max:            284.809618ms

Per-Operation Latency:
  QUERY_NUMERIC (29998 requests):
    Avg: 3.819567ms, P50: 2.840784ms, P95: 7.126519ms, P99: 11.679857ms, Max: 284.809618ms
```

NUMERIC REDIS

KẾT LUẬN:

1. Hiệu suất thao tác Key-Value cơ bản (SET, GET, DELETE)

- Aerospike được tối ưu hóa cho các khối lượng công việc cường độ cao. Hệ thống dễ dàng duy trì thông lượng lên tới 100.000 TPS cho các thao tác GET và DELETE, hoàn thành bài kiểm tra đúng thời gian với độ trễ trung bình và P95 cực thấp. Điểm mạnh lớn nhất của Aerospike là sự ổn định và khả năng giữ vững hiệu suất mà không bị rớt gói tin ở mức tải lớn.
- Redis hoạt động cực kỳ nhanh ở các thao tác cơ bản (như SET) với thời gian phản hồi trung vị (P50) thường thấp hơn Aerospike nhờ kiến trúc thuần in-memory. Tuy nhiên, khi đẩy tải lên ngưỡng cực cao (100.000 TPS với DELETE), Redis bắt đầu có dấu hiệu thắt cổ chai nhẹ, làm tụt TPS hoàn thành và kéo dài thời gian thực thi tổng thể.

2. Khả năng truy vấn phức tạp (QUERY_TAG, QUERY_NUMERIC)

- Redis (với RediSearch) thể hiện sự áp đảo hoàn toàn. Cấu trúc chỉ mục của hệ thống này cho phép xử lý các truy vấn thẻ (Tag) và đặc biệt là dải số (Numeric) với hiệu suất và độ tin cậy vượt trội. Ngay cả khi tải mục tiêu cao gấp 4 lần so với Aerospike trong bài kiểm tra Numeric, Redis vẫn xử lý 100% khối lượng công việc với độ trễ rất thấp, không có bất kỳ yêu cầu nào bị hủy.
- Aerospike bộc lộ điểm yếu chí mạng khi xử lý truy vấn dữ liệu. Hệ thống nhanh chóng rơi vào trạng thái quá tải, dẫn đến thời gian phản hồi trung bình tăng vọt và buộc phải hủy bỏ (abandon) hàng ngàn yêu cầu, ngay cả khi mức tải mục tiêu được thiết lập rất thấp (chỉ 700 TPS cho truy vấn Numeric).

3. Tính ổn định và quản lý độ trễ đuôi (Tail Latency)

- Aerospike quản lý các giới hạn độ trễ tối đa (Max Latency) vô cùng ấn tượng. Trong mọi kịch bản, dù hệ thống xử lý mượt mà hay phải hủy bỏ yêu cầu do quá tải, mức độ trễ cao nhất hiếm khi vượt qua ngưỡng an toàn, giúp hạn chế triệt để các đợt giật lag (spikes) kéo dài cho ứng dụng.
- Redis dễ xuất hiện các "gai" độ trễ ở những tình huống hệ thống bị đẩy đến giới hạn chịu đựng. Dù P95 và P99 duy trì rất tốt, độ trễ tối đa của Redis có thể vọt lên hàng trăm mili-giây (như 362ms ở truy vấn Tag hay 284ms ở truy vấn Numeric).